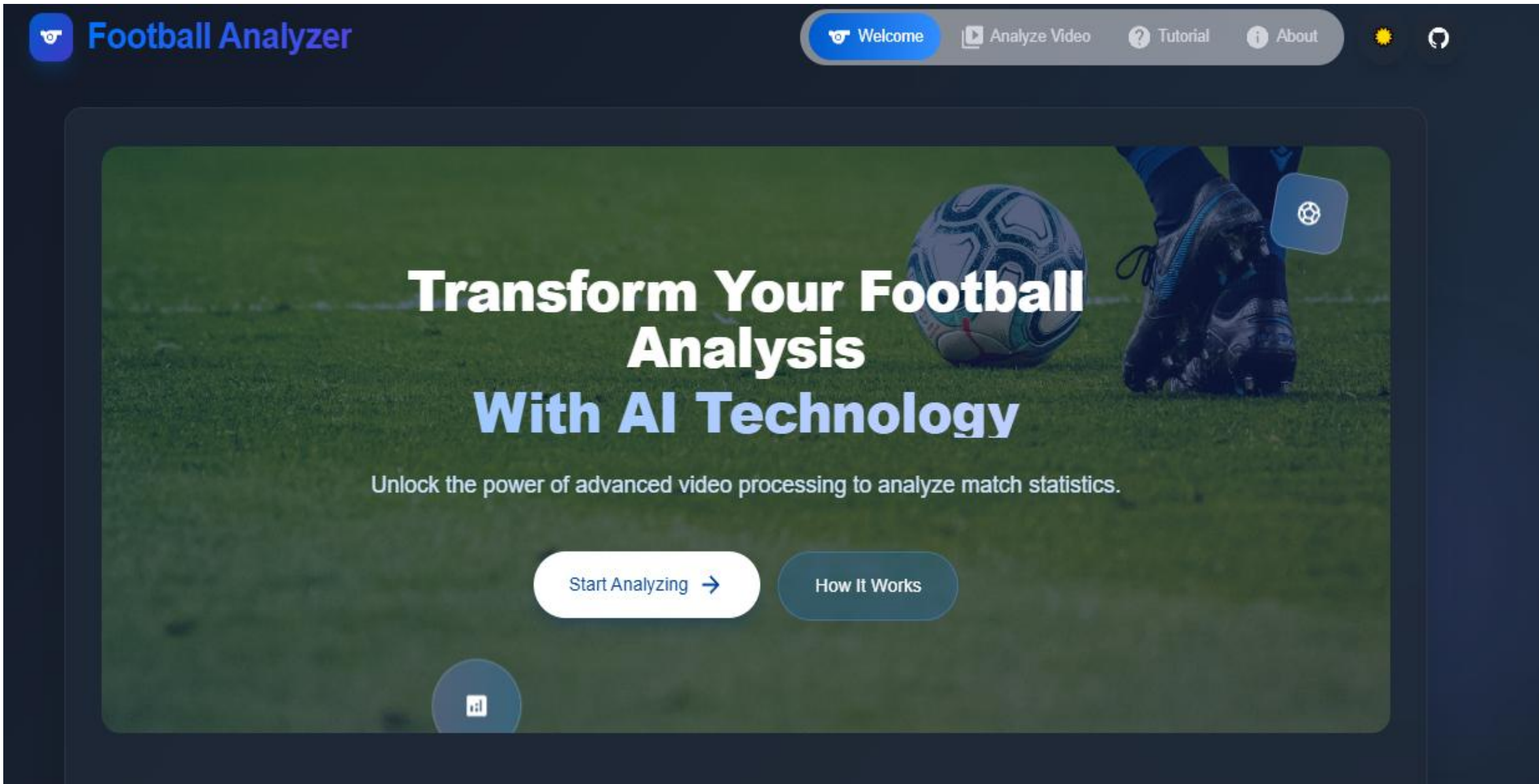


Football Performance Analyzer

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BEng(H) Software and Electronic Engineering





Introduction

The Football Player Analysis (FPA) is an advanced computer vision system that automates the analysis of football matches. It combines modern machine learning techniques with computer vision to provide real-time insights into player movements, team assignment, and game statistics. The system processes video footage to detect players, track the ball, and calculate possession statistics, providing valuable data that was back in the day only available through manual analysis.



Problem Statement

- Traditional football analysis methods face several challenges:
- Manual video analysis is time-consuming and labour-intensive
 - Human analysts can miss critical details in fast-paced game situations
 - Real-time analysis during matches is nearly impossible
 - Consistent and objective data collection is difficult to maintain
 - Detailed statistics require multiple analysts working for hours

Introduction

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Why soccer Analysis?

- Growing up in a football-loving family, the sport has always been more than just a game to me - it's a passion and hobby I can't live without. When it came time to choose a project, I wanted to create something that truly resonated with my interests and would keep me motivated throughout the development process.
- The inspiration came from watching halftime analysis of the Arsenal Brighton game which gave me the idea Why don't I do this?, where analysts use sophisticated tracking systems to break down player movements and game strategies. These systems use small tracking markers to follow players, providing detailed insights into:
 - Player positioning
 - Movement patterns
 - Tactical decisions
 - Performance statistics

This sparked my idea to create the football performance analyzer which could provide similar analysis capabilities, making insights accessible through computer vision and machine learning. The project combines my passion for football with my interest in technology, creating a solution that could revolutionize how we analyze and understand the beautiful game.

Premier League · 31 Aug 24

Full-time



Arsenal

1

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Brighton

Kai Havertz 38'

⊕

João Pedro 58'

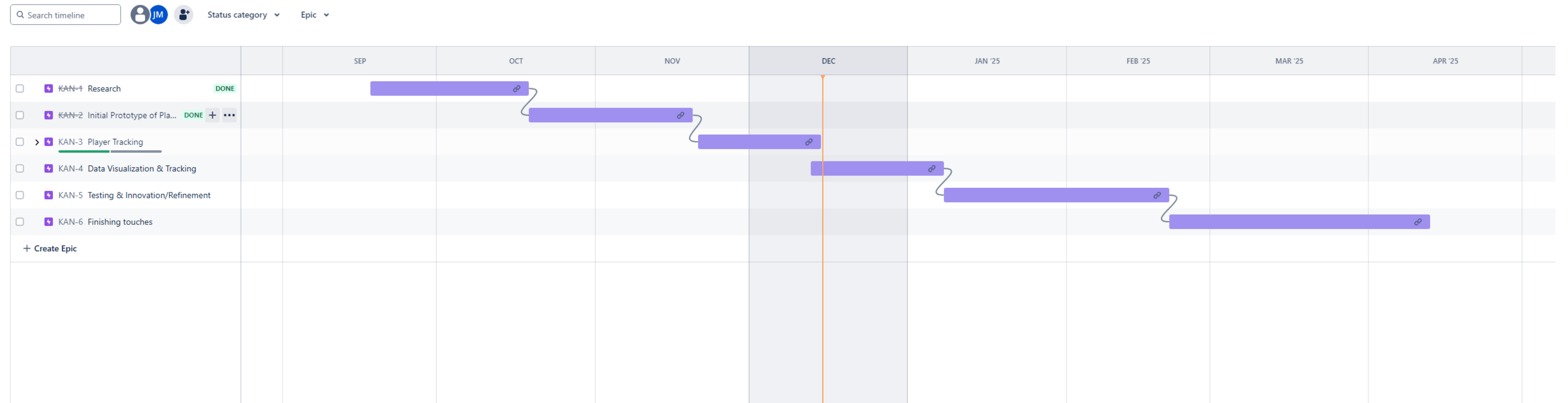
Declan Rice 49'

■

Project Planning

- **Method Used:** Agile Methodology with Jira
- **Key Points:** Managed tasks, timelines, and workflows using Jira
- Organized project into structured phases with clear milestones
- Enabled real-time collaboration and progress tracking
- Logged and resolved issues efficiently
- Visualized progress with Gantt charts and dashboards
- Adapted plans based on feedback and retrospectives
- **TimeLine: Research ✓** (September)
- Initial Prototype ✓ (October)
- Player Tracking ✓ (December; Team Assigner with player tracking Feb–Mar)
- Data Visualization & Tracking ✓ (April)
- **Current Phase:** Finished

Timeline

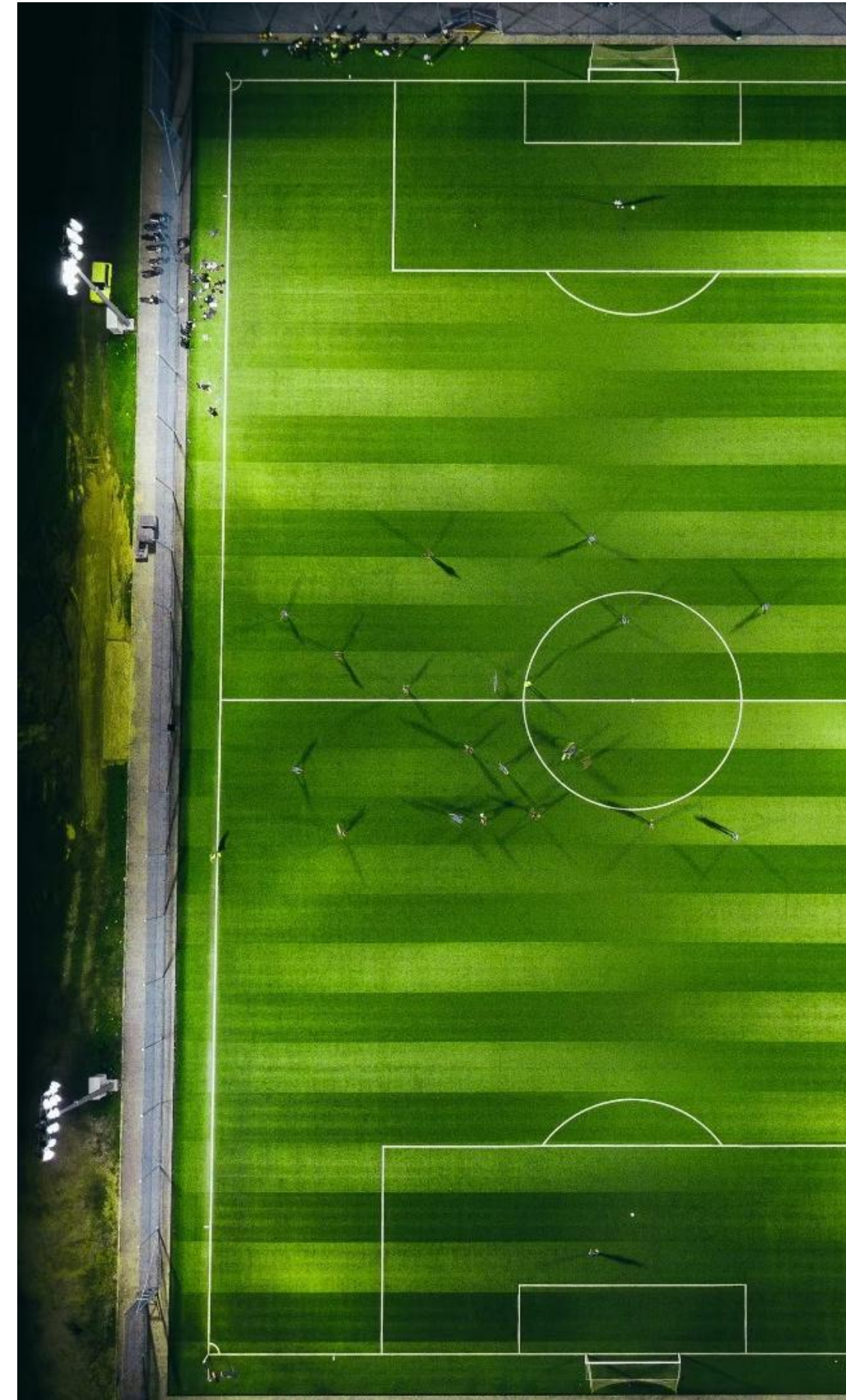


Project overview

Football Player Analysis (FPA) is an advanced system that leverages computer vision and machine learning to automate the analysis of football matches. The platform processes match videos to detect players, assign teams, track the ball, and calculate real-time possession statistics. By providing detailed visualizations and actionable insights, FPA aims to make professional-level football analytics accessible to coaches, analysts, and enthusiasts.

What's Unique About FPA?

- Fully automated, end-to-end football analysis
- Real-time player and ball tracking with YOLOv8 Roboflow Model
- Possession done using the Euclidean Theory
- Smart team assignment using colour clustering
- Instant, interactive results for users
- Scalable and parallel processing with RabbitMQ
- Modern, extensible tech stack and architecture



Technologies Used

Frontend

- React.js**: Built a single-page app with custom components for video upload, progress tracking, and result visualization.
- Material UI**: Provided a responsive layout with customized components tailored to football analysis.
- Tailwind CSS**: Applied utility-first styling for rapid and consistent UI customization.
- Axios**: Managed all HTTP communication, including file uploads, status polling, and result retrieval.
- Framer Motion**: Added smooth animations for enhanced user experience.

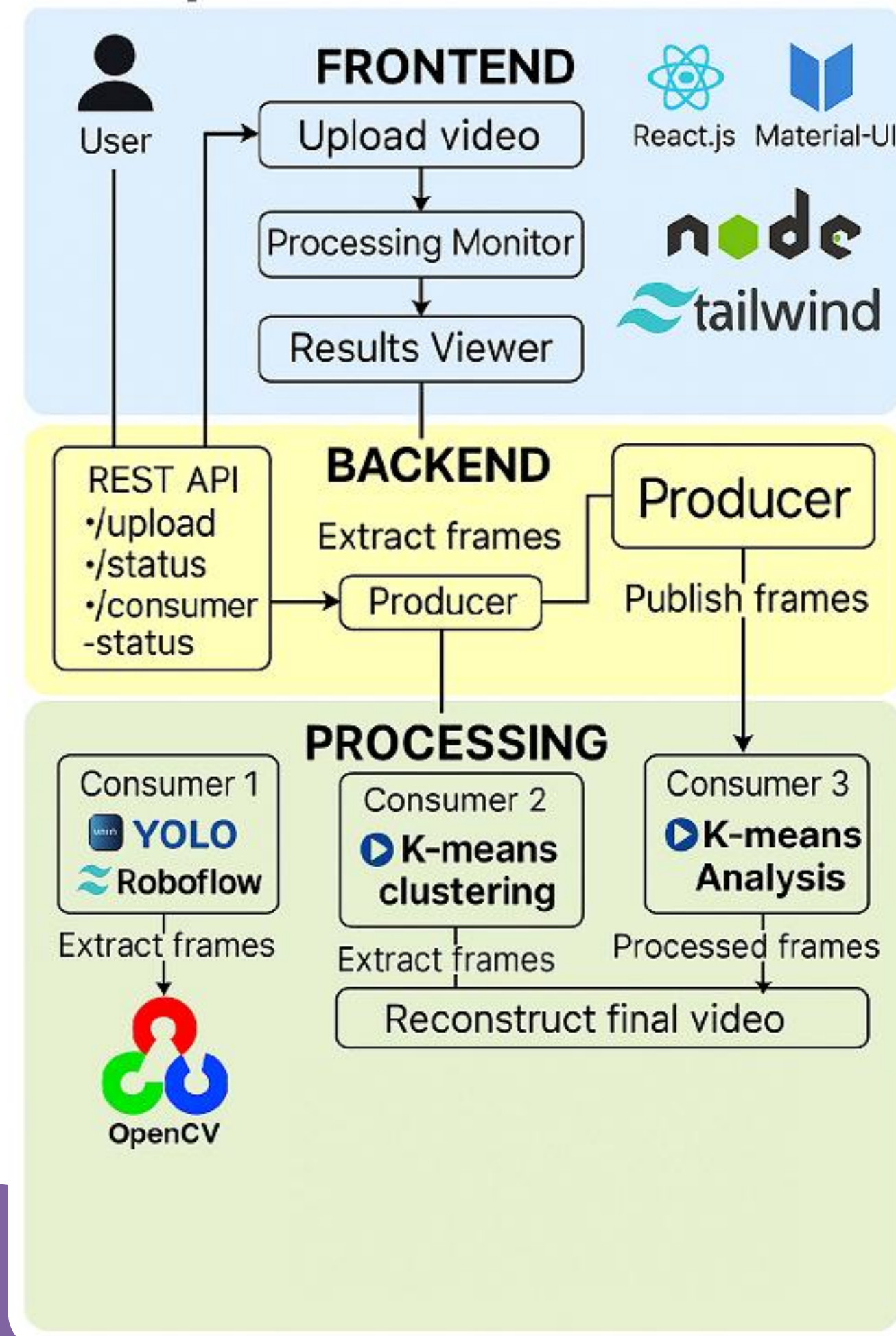
Backend

- Node.js**: Developed a backend server to process API requests and coordinate video processing.
- Express.js**: Created REST API endpoints for uploading videos, checking status, and retrieving results.
- Multer**: Handled and managed incoming video uploads from the frontend.

Video Processing

- OpenCV**: Captured video frames, drew bounding boxes and team labels, and produced the final processed video.
- Roboflow API**: Used pre-trained models for player, referee, and ball detection. Initially considered training a custom model, but on advice from Paul Lennon, adopted Roboflow for higher accuracy and fewer false detections (e.g., fans mistaken as players).
- NumPy**: Performed efficient image array operations and colour distance calculations for team assignments.
- scikit-learn**: Applied K-means clustering to group detected player jersey colours and assign teams.
- pika (RabbitMQ client)**: Implemented producer-consumer scripts to enable scalable and efficient distributed frame processing.





System architecture

Frontend Application:

- User-friendly web interface (React, Material-UI, Tailwind CSS) for video upload, progress monitoring, and results visualisation.

Backend Server:

- Node.js/Express server that manages uploads, coordinates processing, and acts as the central hub for data flow.

Video Processing Engine:

- Python-based engine for player detection (YOLOv8), team assignment (K-means), and possession analysis, distributed across multiple consumers.

Euclidian Distance theory

How Euclidean Distance is Used in Our Code

In the Football Player Analysis (FPA) system, the Euclidean Distance formula is a core component for determining ball possession in each video frame.

How it works in our code:

Detection:

For every frame, the object detection model (YOLOv8 via Roboflow) identifies all players and the ball, outputting their x, y coordinates.

Distance Calculation:

The code calculates the Euclidean distance between the ball and every detected player using the formula in the bottom right corner

Possession Assignment:

The player with the smallest distance to the ball is considered to be in possession for that frame. The code then assigns possession to that player's team.

Statistics Generation:

By repeating this process for every frame, the system accumulates possession statistics for each team throughout the match.

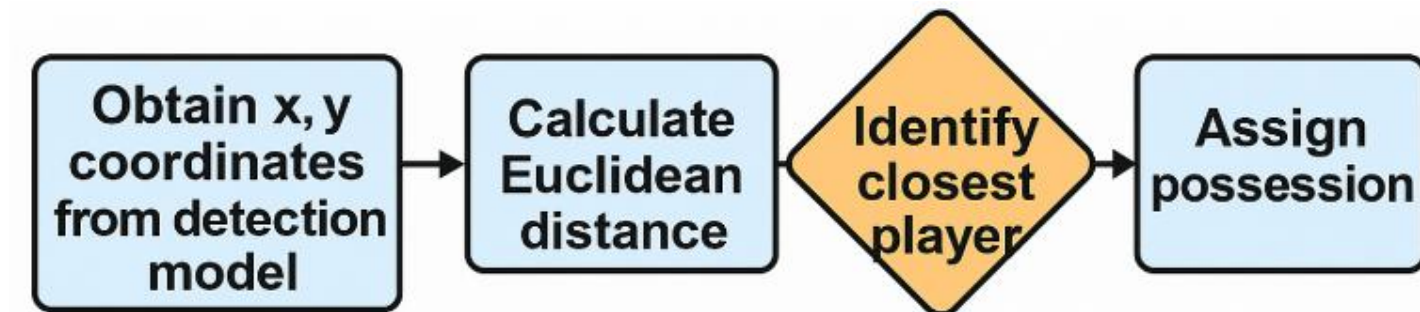
Why this is important:

This method provides an objective, frame-by-frame calculation of which team has possession, enabling accurate and automated match analysis.

The extracted data is also used for further analytics, such as player tracking, team formation analysis, and visualizations.

In summary:

The Euclidean Distance calculation is the mathematical backbone that allows our code to automatically and accurately determine ball possession, making advanced football analytics possible without manual intervention.



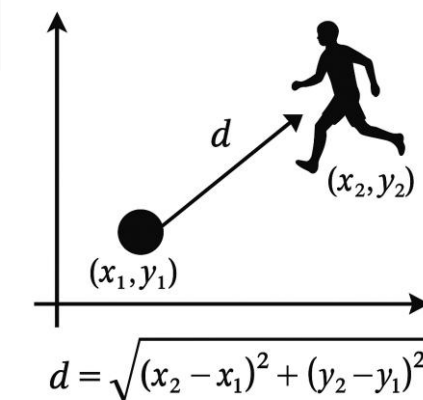
Euclidean Distance Formula

Consider two points (x_1, y_1) and (x_2, y_2) in a 2-dimensional space; the Euclidean Distance between them is given by using the formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Where,

- d is Euclidean Distance,
- (x_1, y_1) is the Coordinate of the first point,
- (x_2, y_2) is the Coordinate of the second point.



Roboflow Model

Model Specifications:

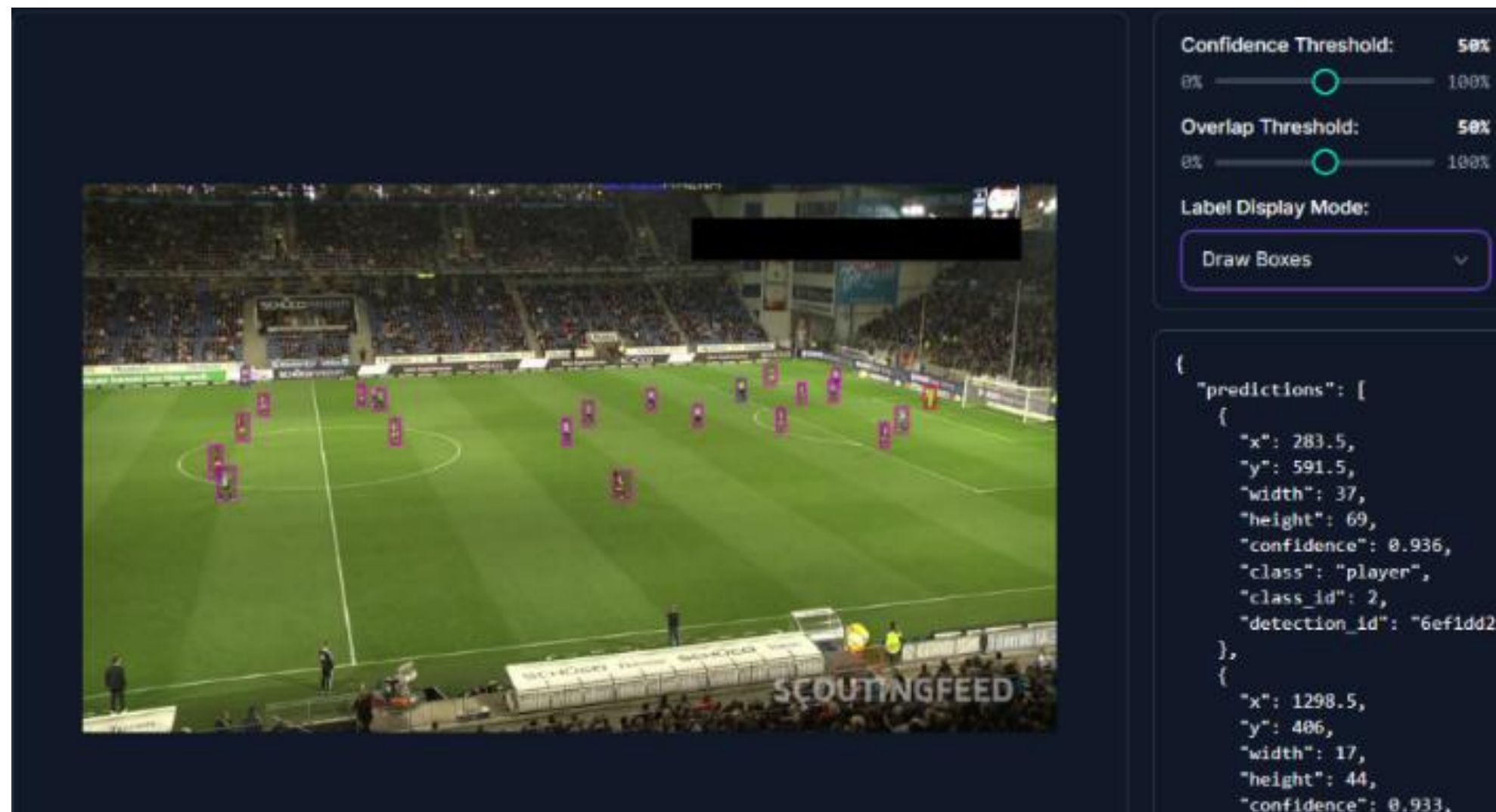
- Specifically designed for football/soccer analysis
- Optimized for real-time player and ball tracking

Key Capabilities:

- Accurately detects players, referees, and the ball in each frame
- Performs well even with partially occluded players
- Enables automatic team assignment

Technical Implementation:

- Integrates with K-means clustering for team jersey identification
- Allows for real-time possession tracking and statistics generation
- Provides foundation for advanced analytics in the processing pipeline



Key features

- Automated Player Detection:**

Detects all players on the pitch in each video frame using a custom-trained YOLOv8 model via Roboflow.

- Ball Detection and Tracking:**

Identifies and tracks the ball's position throughout the match for accurate analysis.

- Team Assignment via Colour Clustering:**

Automatically assigns players to teams using K-means clustering based on jersey colours.

- Real-Time Possession Analysis:**

Calculates which team has possession of the ball in each frame using the Euclidean distance between players and the ball.

- Parallel and Scalable Processing:**

Utilizes RabbitMQ to distribute video frame processing across multiple consumers, enabling faster and scalable analysis.

- Interactive Web Interface:**

User-friendly frontend (React, Material-UI, Tailwind CSS) for uploading videos, monitoring progress, and viewing results.

- RESTful API Backend:**

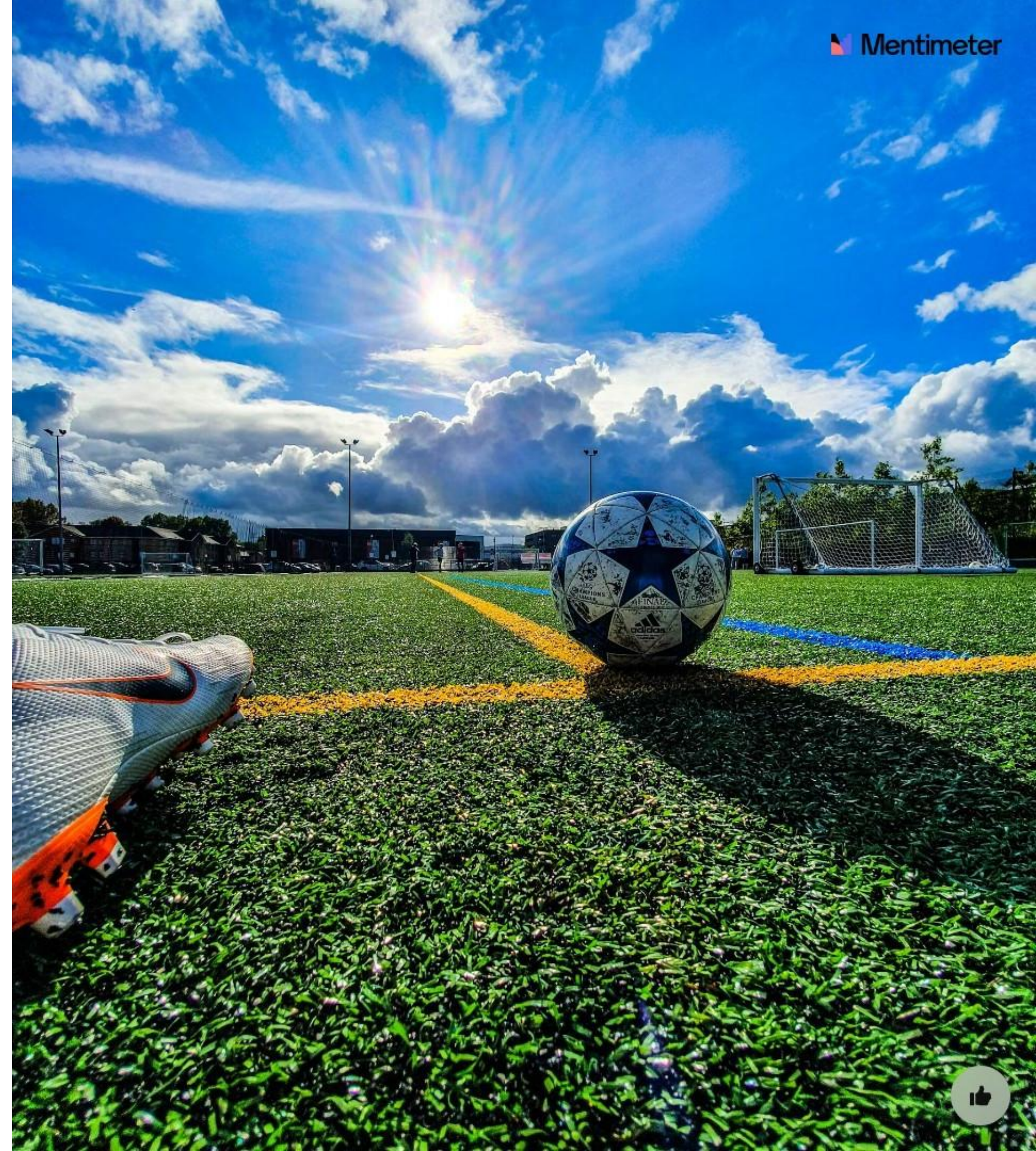
Node.js/Express backend manages video uploads, processing coordination, and serves analysis results.

- Custom Visualization:**

Generates processed videos with overlays (bounding boxes, team colours, possession stats) for clear, actionable insights.

- Seamless Integration:**

Integrates Python, OpenCV, and Roboflow for efficient computer vision processing



Challenges and Solutions

Player Occlusion

- **Challenge:** Players frequently occluded each other, causing detection failures.
- **Solution:** Introduced temporal consistency checks to interpolate missing player positions.

Team Assignment Consistency

- **Challenge:** Team assignments occasionally switched between frames.
- **Solution:** Added persistence tracking and threshold-based reassignment to stabilize team labelling.

Processing Efficiency

- **Challenge:** Sequential frame processing was too slow for practical deployment.
- **Solution:** Rebuilt the system using distributed processing with message queuing (RabbitMQ).

Model creation

- **Challenge:** create my own model huge difference in accuracy it was so bad the whole stadium was being detected
- **Solution:** Roboflow

Possession Calculation

- **Challenge:** Attempted updating possession every 50 frames to reduce jitter, but this caused update failures.
- **Solution:** Reverted to updating possession every frame, accepting slight jitter for consistent behaviour.



Web Interface

- Modern, Responsive Design:** Built with React.js, styled using Material UI and Tailwind CSS for a clean and consistent look.
- Simple Upload Workflow:** Allows users to easily upload football match videos for analysis.
- Real-Time Processing Feedback:** Displays processing status updates via a progress component that monitors backend activity.
- Animated User Experience:** Integrated Framer Motion for smooth animations during uploads, status updates, and result displays.



Results and evaluation

- The Football Performance Analyzer system successfully generates two major outputs:

- Possession Statistics:**

Real-time calculation of which team controls the ball throughout the match, based on frame-by-frame Euclidean distance analysis. Final statistics are presented as percentages for each team's total possession time.

- Annotated Final Video:**

A fully processed match video that shows players detected with bounding boxes, team colours assigned, and possession tracking visualized live during play. This video provides clear, visual feedback of the system's tracking, team assignment, and analysis capabilities.





Future Enhancements

• 1. Enhanced Player Tracking

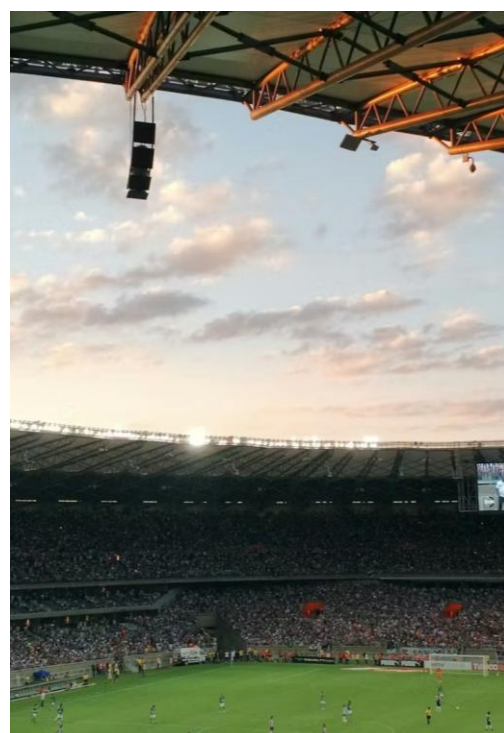
- Implement deep learning-based re-identification models.
- Introduce jersey number recognition to maintain player identities.
- Use trajectory prediction to track players during occlusions.
- Improve goalkeeper and player team assignment accuracy.

• 2. Additional Statistical Analysis

- Add pass detection and success rate metrics.
- Detect shots and calculate conversion rates.
- Generate heat maps of player movements.
- Perform formation and tactical analysis.
- Measure physical metrics (e.g., distance covered, sprints).

• 3. Real-Time Processing Improvements

- Develop real-time player tracking for continuous identity preservation.
- Integrate jersey number recognition for identity consistency.
- Enhance trajectory prediction models for smoother tracking.
- Enhance the possession for smoother transition of values.





Lessons learned

Efficient Video Processing is Critical

Optimizing frame extraction, annotation, and reconstruction greatly improved overall system performance.

Complexity of Computer Vision in Sports

Handling occlusions, jersey similarities, and ball detection proved to be much more challenging than initially expected.

Value of Distributed Processing

Using a producer-consumer model with RabbitMQ enabled scalable and faster processing across multiple CPU cores.

Importance of Error Handling in Production Systems

Implementing robust error recovery, graceful degradation, and comprehensive logging was essential for maintaining system reliability during unexpected conditions.



Conclusion

Conclusion

Automated Football Analysis

The FPA system uses machine learning and computer vision to deliver fast, cost-effective, and accurate football video analysis without human intervention.

Open-Source, User-Friendly Design

A web-based, distributed system that enables efficient processing even on standard consumer hardware.

Broad Accessibility

By offering an open platform, FPA makes professional-quality analytics available to amateur clubs and individual coaches.

Future Potential

Planned enhancements include real-time analytics, expanded statistics, and advanced player tracking to further improve football performance insights.



Q&A